

The Greeks

Delta, Gamma, Theta & Vega Explained — Lesson Explainer

The Concept

Before expiration, an option's premium moves constantly, not just because of the stock price. The Greeks each isolate how sensitive an option's price is to one specific factor at a time.

Delta: change per \$1 move in the stock. Gamma: how much delta itself changes ('the delta of delta'). Theta: value lost purely from one day passing. Vega: sensitivity to shifts in implied volatility.

Together they answer: if the stock moves, if time passes, or if volatility expectations shift, what happens to this position? Watching only the stock price ignores at least three of the four forces at work.

Illustrative Example: Reading a Position's Greeks

Greek	Value	What It's Saying
Delta	+0.55	Gains ~\$0.55 per \$1 the stock rises
Gamma	0.04	Delta itself rises ~0.04 per \$1 stock move
Theta	-0.08	Loses ~\$0.08 of value per day, all else equal
Vega	+0.12	Gains ~\$0.12 if implied volatility rises 1 point

Every day this option is held, theta quietly erodes ~\$0.08 of value even with no stock move - a cost only a directional move, an accelerating move, or a rise in volatility expectations can offset.

5 Things to Know About the Greeks

1. Delta also approximates probability of expiring in the money, as a rough rule of thumb.
2. Gamma is highest near the strike, close to expiration - delta can shift fastest here.
3. Theta accelerates as expiration nears - decay isn't linear, it speeds up in the final weeks.
4. Vega matters most for longer-dated options - more time means more sensitivity to volatility.
5. The Greeks interact, not operate in isolation - real P&L reflects all four moving together.

4 Things to Check Before Holding an Options Position

- Check your delta exposure - know roughly how much your position moves per \$1 in the underlying.
- Respect theta as a buyer - a long option bleeds value every day, even if the stock stays flat.
- Watch vega around known events - implied volatility often drops sharply after earnings.
- Mind gamma near expiration - directional exposure can change quickly in the final days.

Worth knowing:

this explains the Greeks using illustrative numbers - it isn't personalized financial advice, and no position or value here is a recommendation. The Greeks are theoretical sensitivities from a pricing model, not guarantees of how an option will actually behave.

Quiz — Answer the questions, then check the key on the next page

1. What does Delta measure?

- A. How much value an option loses per day
- B. How much the option's price changes for a \$1 move in the underlying stock
- C. The option's strike price
- D. How many days remain until expiration

2. What does Theta represent?

- A. Sensitivity to the stock price
- B. Sensitivity to implied volatility
- C. Time decay - value lost purely from one day passing, all else equal
- D. The maximum possible loss on the position

3. In the illustrative example, what happened to the option's value every day it was held, even with no stock move?

- A. It lost about \$0.08 of value from theta decay
- B. It gained value automatically
- C. Its strike price changed
- D. Delta dropped to zero

4. According to this lesson, when is Gamma typically highest?

- A. Only on options with no expiration date
- B. When the stock price is far from the strike
- C. Immediately after the option is purchased, regardless of other factors
- D. Near the strike price, close to expiration

5. What does Vega measure?

- A. The option's delta squared
- B. The number of shares per contract
- C. Sensitivity to changes in implied volatility
- D. The commission cost of the trade

Answer Key

1. What does Delta measure?

Correct answer: B. How much the option's price changes for a \$1 move in the underlying stock

Delta measures how much the option's price is expected to change for a \$1 move in the underlying.

2. What does Theta represent?

Correct answer: C. Time decay - value lost purely from one day passing, all else equal

Theta measures how much value an option loses purely from the passage of time.

3. In the illustrative example, what happened to the option's value every day it was held, even with no stock move?

Correct answer: A. It lost about \$0.08 of value from theta decay

Theta of -0.08 means the option loses about \$0.08 of value each day purely from time passing.

4. According to this lesson, when is Gamma typically highest?

Correct answer: D. Near the strike price, close to expiration

Gamma tends to be highest near the strike price as expiration approaches.

5. What does Vega measure?

Correct answer: C. Sensitivity to changes in implied volatility

Vega measures how much an option's price changes if volatility expectations shift, stock price unchanged.